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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services: Design and Experimental Characterization

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Master's in Telecommunications and Computer Engineering

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Ph.D. Professor Pedro Ramos, Assistant Professor,
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Month, Year

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Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

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Keywords: *Keywords; separated; by; semicolons*

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List of Acronyms

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CHAPTER 1

Introduction

1.1. Context and Motivation

1.2. Objectives

1.3. Research Questions

1.4. Methodology Overview

1.5. Contributions

1.6. Document Structure

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CHAPTER 2

Literature Review

2.1. Review Method

2.2. Edge Computing Architectures

2.3. SDN and Programmable Infrastructure

2.4. Auto-Scaling and Elasticity Mechanisms

2.5. Metadata-Driven Data Placement and Replication

2.6. Document-Oriented Databases for Edge Environments

2.7. Synthesis

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CHAPTER 3

System Architecture

3.1. Design Principles

The proposed architecture is implemented on an sdn network, where all the decision making and monitoring done for each network is made by the controller who owns and controls the network.

3.2. Architecture Overview

3.3. Telemetry Pipeline

3.4. Routing and Backend Selection

3.5. Elasticity and Data Gravity Tiers

3.6. Experimental Workload

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CHAPTER 4

Methodology

- 4.1. Research Approach
- 4.2. Experimental Design
- 4.3. Variables
- 4.4. Measurement Instrumentation
- 4.5. Experiment Procedure
- 4.6. Repetition and Statistical Treatment
- 4.7. Validity and Reliability

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CHAPTER 5

Implementation

- 5.1. SDN Controller
- 5.2. Edge Service and Storage
- 5.3. Network Infrastructure
- 5.4. Analysis Toolchain
- 5.5. Implementation Validation

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CHAPTER 6

Results and Discussion

- 6.1. RQ1 — Telemetry Freshness and Delivery Cadence
- 6.2. RQ2 — Metadata-Aware Backend Selection
- 6.3. RQ3 — Data Locality Strategies
- 6.4. Cross-RQ Synthesis

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CHAPTER 7

Conclusions and Future Work

7.1. Summary of Findings

7.2. Contributions Revisited

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